Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement
Recommendations to harden the organization's network after the inspection done as a result of our recent data breach and the discovery that the network was in need of more effective prevention measures are: 1. Password policies - methods to salt or hash passwords rather than requiring overly complex passwords or enforcing frequent changes to passwords, employees may no longer share passwords and the admin password may no longer be the default password for security purposes, making it more difficult for a brute force attack. The security team will be working with employees to make sure they each have their updated password and with the admin team to make sure they have a password that is not the default password. 2. Firewall maintenance - the firewall rules need to updated in response to the recent event to prevent abnormal network traffic from having the ability to enter the network, thus preventing DDoS attacks. The rules can be updated regularly by the security team in response to any future events and to prevent any future events from occurring. 3. Multifactor Authentication [MFA] - Requires users to verify their identity in 2 or more ways to access a system or network. Protects against brute force attacks and similar security events. The security team will set this system up and maintain it regularly after the initial setup to prevent future events of this kind.

Part 2: Explain your recommendations
After inspecting the organization's network, four major vulnerabilities were discovered that needed to be addressed. 1. Employees share passwords. 2. The admin password for the database is set to default.

3. The firewalls do not have rules in place to filter traffic coming in and out of the network.
4. Multifactor authentication [MFA] is not in use.

   If no action is taken to address these vulnerabilities, the organization is at risk of experiencing another data breach or other attacks in the future.